SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Sol:

Given

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

Step 1: Identify the subnet

A /26 subnet has a block size of 64.

Subnets are:

Subnet	Address Range
192.168.10.0/26	0–63
192.168.10.64/26	64–127
192.168.10.128/26	128–191
192.168.10.192/26	192–255

Since 192.168.10.65 lies between 64–127, it belongs to

Subnet = 192.168.10.64/26

Step 2: Network Address

192.168.10.64

Step 3: Broadcast Address

192.168.10.127

Step 4: Valid Host Range

192.168.10.65 – 192.168.10.126

Step 5: Verify IP Address

IP = 192.168.10.65

✓ Valid Host

Step 6: Verify Gateway

Gateway = 192.168.10.1

This gateway belongs to subnet

192.168.10.0/26

But the PC belongs to

192.168.10.64/26

❌ Gateway is incorrect.

Configuration Error

- IP Address → Correct
- Gateway → Wrong subnet

Optimized Configuration

Setting	Value
IP Address	**192.168.10.65**
Subnet Mask	**255.255.255.192**
Default Gateway	**192.168.10.65's router interface, e.g. 192.168.10.126 (or 192.168.10.66, depending on router design)**

Number of Usable Hosts

Formula

$$\mathbf{2^{(32-26)} - 2}$$
$$\mathbf{2^6 - 2 = 64 - 2 = 62}$$

Answer = 62 usable hosts

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Sol:

Device A

2001:db8::ff00:42:8329/64

Expanded

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:db8:0:0:1::1/64

Expanded

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

Device A

2001:db8:0:0::/64

Device B

2001:db8:0:0::/64

Same Subnet?

Yes.

Both have

2001:db8:0:0::/64

✔ Same subnet.

Why Communication Fails?

The subnet prefix matches, so the failure is not due to addressing. Likely causes include:

- Incorrect interface configuration
- Duplicate address
- Interface down

- Firewall or ACL blocking traffic

Optimized IPv6 Addressing

Device A

2001:db8:0:0::10/64

Device B

2001:db8:0:0::20/64

Number of Host Addresses

A /64 network has

$\mathbf{2^{64}}$

host addresses.

Answer = 18,446,744,073,709,551,616 addresses

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

<u>Task</u>

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Sol:

Network

192.168.1.0/24

/26

Network

192.168.1.0

Broadcast

192.168.1.63

Host Range

192.168.1.1 – 192.168.1.62

Usable Hosts

62

/27

Network

192.168.1.64

Broadcast

192.168.1.95

Host Range

192.168.1.65 – 192.168.1.94

Usable Hosts

30

/28

Network

192.168.1.96

Broadcast

192.168.1.111

Host Range

192.168.1.97 – 192.168.1.110

Usable Hosts

14

Unused IP Addresses

If departments require fewer hosts than allocated, the remaining usable IPs are unused. For example:

- /26 → 62 usable
- /27 → 30 usable
- /28 → 14 usable

Unused IPs = Allocated usable hosts − Actual hosts used.

Inefficient Allocation

Using /26 for a small department wastes addresses.

Optimized VLSM

Allocate subnets based on department size:

Department	Subnet
Large	/26
Medium	/27
Small	/28

This minimizes wasted IP addresses.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Sol:

Expanded Address

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

2001:db8:0:0::/64

Same Subnet

✓ Yes

Optimized Addresses

Device A

2001:db8:0:0::10/64

Device B

2001:db8:0:0::20/64

Available Hosts

2^{64}

= **18,446,744,073,709,551,616**

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Sol:

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP

192.168.2.100

Longest Prefix Match

Destination

192.168.2.100

belongs to

192.168.2.0/24

✓ Matching Route

192.168.2.0/24

Next-Hop Router

10.0.0.3

Destination Subnet

192.168.2.0/24

✓ Correct

Possible Routing Errors

- Wrong next-hop IP
- Interface down

- Missing route
- Incorrect subnet mask
- Router connectivity issue

Optimized Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

(Ensure all next-hop interfaces are reachable and active.)

If No Matching Route Exists

The router uses the

Default Route → 10.0.0.1

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation